CSU IREC Team Composite Layup Analysis for Subscale and Full-Scale Airframe

The calculations below showcase the thickness of the layup, the mass and the force to be analyzed in ANSYS:

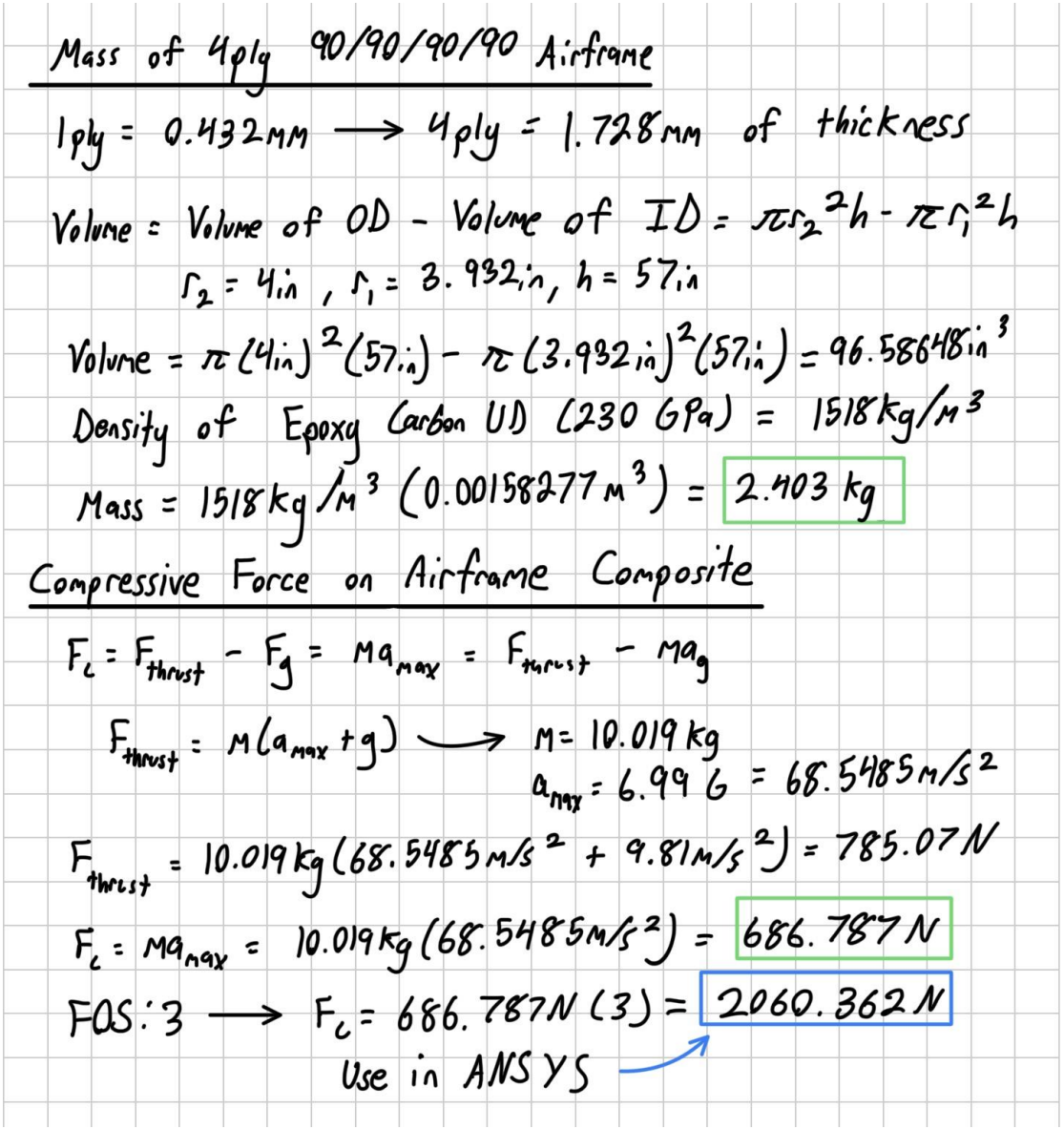


Figure 1: Composite Calculations

With each layer of the 4-ply laminate being 0.432mm, the total thickness ends up being 1.728mm, This dictates a mandrel diameter of 3.932in to allow an outside diameter of four inches for the airframe. Using the volume calculation from the outside volume minus the mandrel volume, the mass of the airframe as a whole was determined to be 2.403kg. After plugging this into our openrocket model, we got a max acceleration of the rocket to be 6.99G. Using that acceleration, the compressive force experienced by the airframe with a factor of

safety of 3 was determined to be 2060.362N. This let us use FEA to determine which layup pattern would have the least deformation under this force.

Test 1: 30/60/30/60

Using a 4-ply laminate with the ply angles of 30, 60, 30, and 60 degrees respectively, and applying a fixed support on one end and a compressive force of 2060.362N on the other, the max deformation of this laminate was found to be 0.39289mm

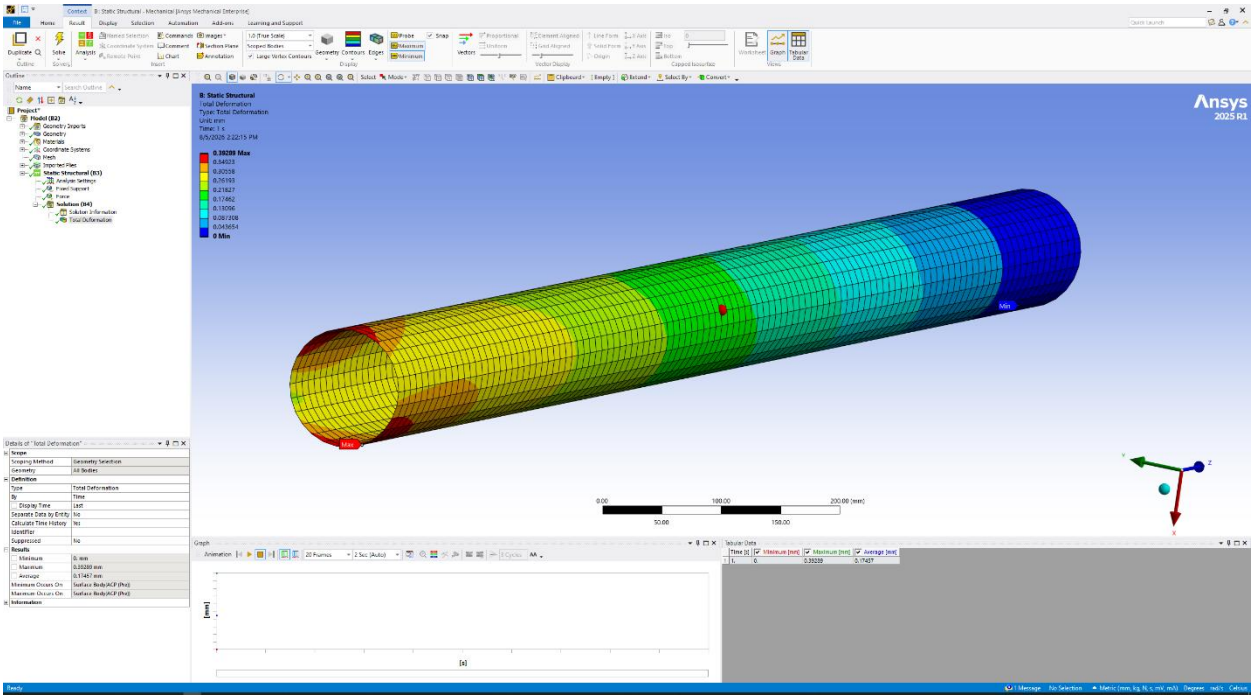


Figure 2: 30/60/30/60 Airframe Laminate Deformation Test

Test 2: 90/90/90/90

Using a completely hoop-dominant layup and the same boundary conditions as the previous test, the maximum deformation of the laminate was found to be 0.18195mm.

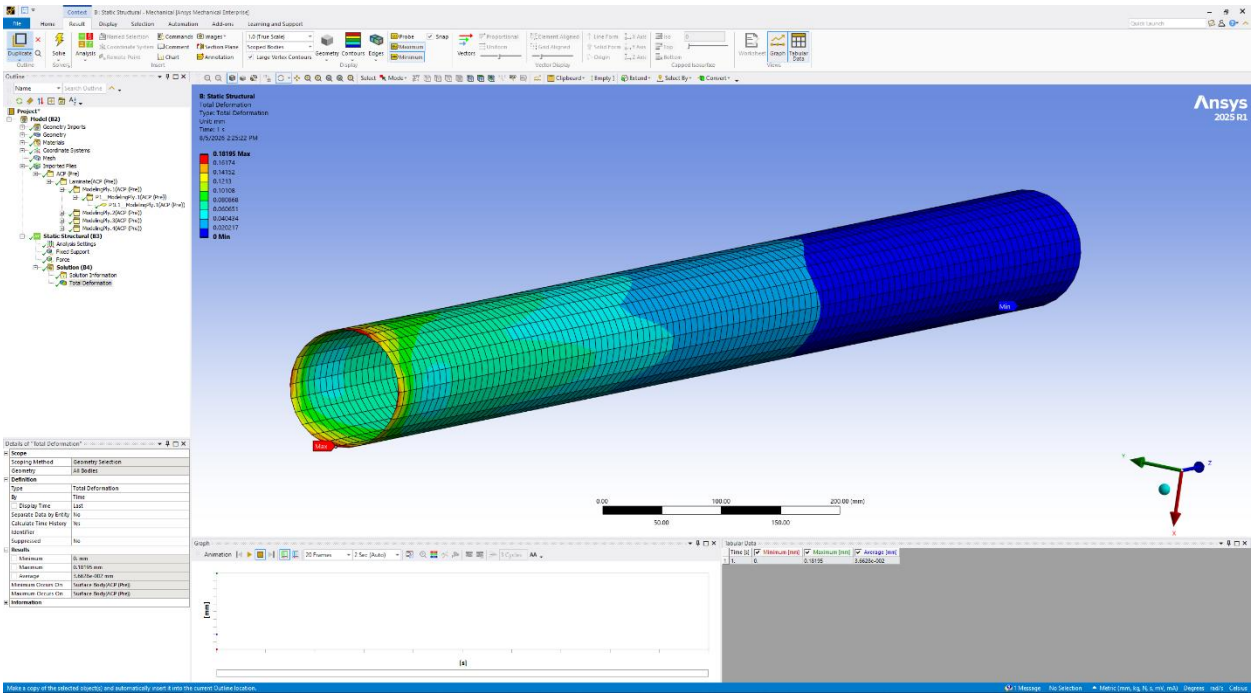


Figure 3: 90/90/90/90 Airframe Laminate Deformation Test

Test 3: 90/55/-55/90

Using a mixed layup of hoop-dominant and cylindrical 55/-55 degree angles, the maximum deformation of this layup was found to be 0.11589mm

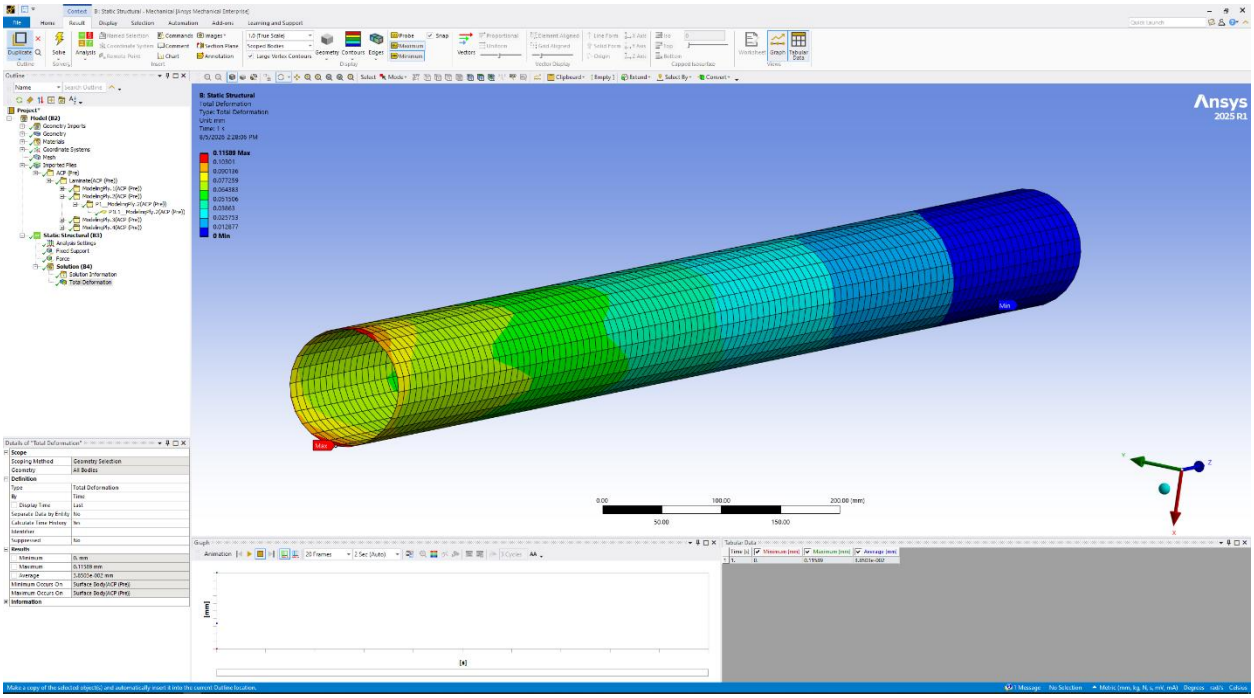


Figure 4: 90/55/-55/90 Airframe Laminate Deformation Test

Using the above tests, and other non-documented results, the most optimal layup for our purposes is the 90/55/-55/90 laminate since it has the least deformation under load. This layup will be used for the full-scale model from Ability Composites. For our subscale, the 90/90/90/90 layup will be used as we can only get a hand layup for our subscales targeted launch. While not as sturdy, the layup deals with deformation under load well and is significantly easier to hand layup on mandrel than the helical 55/-55 angles.